COSKELETON COMMUTES WITH PRODUCTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: self-contained given the adjunction of Lemma lemma-existence-cosk; no omitted step.

Editorial note. The label lemma-cosk-product names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Coskeleton functor. Same unit, Lemma lemma-existence-cosk. The coskeleton functor cosk_n with its adjunction is the one constructed in Lemma lemma-existence-cosk.

Lemma 0.1. *Let U, V be n-truncated simplicial objects of a category $\mathcal{C}$. Then*

$$cosk_n(U \times V) = cosk_n U \times cosk_n V$$

whenever the left and right hand sides exist.

Proof. Let W be a simplicial object. We have

$$\begin{aligned} \mathrm{Mor}(W, \mathrm{cosk}_n(U \times V)) &= \mathrm{Mor}(\mathrm{sk}_n W, U \times V) \\ &= \mathrm{Mor}(\mathrm{sk}_n W, U) \times \mathrm{Mor}(\mathrm{sk}_n W, V) \\ &= \mathrm{Mor}(W, \mathrm{cosk}_n U) \times \mathrm{Mor}(W, \mathrm{cosk}_n V) \\ &= \mathrm{Mor}(W, \mathrm{cosk}_n U \times \mathrm{cosk}_n V) \end{aligned}$$

The lemma follows. □

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